

technology would recede into the background of our lives. He called it the age of calm computing and enumerated some rules for that too: "The purpose of a computer is to help you do something else. The best computer is a quiet, invisible servant. The more you can do by intuition, the smarter you are; the computer should extend your unconscious. Technology should create calm."

These, we could say, are the golden rules of ambient electronics design too.

Ambient electronics should ideally be embedded, context-aware, personalised, adaptive and anticipatory systems that are so much a part of your environment that you do not even realise their existence. These devices can be customised to your requirements; their innate intelligence enables them to respond in different manners depending on your mood and needs, and even anticipate your desires and react to them without requiring specific instructions. Their interfaces should be human-centric and unobtrusive.

Piling the stones one over the other

We may not be there yet, but the trends are in favour of ambient electronics becoming a truism. "There are embedded solutions coming along with multifunctional gadgets, services with integrated features, options and so on. We can 'feel' the ambient electronics, amongst others, in lighting automation (lights turning off and on automatically); water tap automation; fire detection and corrective actions; tracking of animals, traffic, materials, items; traffic signals changing based on vehicle presence/absence; intelligent automobiles, including interiors; awareness of empty parking slots; remote monitoring of machines; etc," says Mahesh.

Shikawa adds a few more examples such as smart meters, audio systems that turn the volume up or down depending on the ambient noise or activity, and energy usage profilers. He observes: "Showing the energy use

Key characteristics of ambient electronics

- Unobtrusive
- Adaptable
- Secure
- Human-centric (especially the interfaces)
- Context-aware
- Personalised
- Anticipatory
- Highly network-capable
- Calm

age profile to residents psychologically pressurises them to save energy. Such energy use visualisation has turned out to be a very good motivator for people to save energy in Japan."

From a quick look, it is evident that the base technologies for ambient electronics are relatively mature.

With nanotechnology, smart devices and sensors, it is possible to make unobtrusive hardware. Interoperability, networking technologies, service-oriented architecture and Web 2.0 have made seamless communications and integration of systems possible.

"New sensor technologies, or rather new sensor module technologies that combine existing sensors and new, cheap and yet powerful computer/communication modules so that the data can be collected and analysed by a management computer, and the combination of actuators with home LANs make it very easy to control various devices using computers inside residential housings," explains Ishikawa.

Service discovery, auto-configuration and end-user programmable devices have made it possible for devices to network in an *ad-hoc* and dynamic, yet controllable fashion. Self-testing and self-repairing systems, network security solutions, etc have made devices and networks more secure and reliable.

Human-centric computer interfaces—perhaps the most challenging and important prerequisite for ambient electronics—are also being made possible through intelligent agents, multi-modal interaction and context-

awareness.

"The ambient electronics challenge is no longer about finding out new technologies. All the technologies needed are already there, stabilised and matured. It is more about finding ways to integrate the different technologies in order to realise the solutions. In other words, it is more about 'solutioneering' than basic research," comments Mahesh.

"If there is one big breakthrough that could make it all happen, it would be related to embedding intelligence and context awareness (context sensitivity) into the ecosystem, infrastructure and gadgets. This can be realised by embedding wireless, sensor, actuator, control and network technologies—that will include embedded electronics, embedded software, back-end processing and user interfaces. This is because these can be part of any requirement, ecosystem and solution; are standards independent; and come in extremely small footprints."

From dream to reality

Today, we see sleek, well-designed laptops in various shapes, sizes and designs (including wood-panelled ones) that match your home or office décor. We see lighting automation solutions, magic eyes in air-conditioners, doors that open automatically when someone enters and many other products that, to a small extent, merge with the environment. This is just the start of the beginning.

In the future, we will see lights and air-conditioners that turn on, off or dim based on conditions; switchless environments (with wireless and mobile switches); remote and automatic utility metering (electricity and water); electrical/electronic gadgets that are remotely controlled and monitored; electric outlets that are monitored, measured and controlled over the 'cloud;' solutions activated by ambient conditions like sound, image, light, temperature, smell and fluid levels; and many more exciting products.

And when we see these, we will